ACT SUBJECT TEST & DIGITAL SAT MATH COURSES

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ACT2 • ACT SUBJECT TEST MATH LEVEL 1 • PRACTICE EXAM

ACT2 Exam 6

Full Practice Test • Test #6

Radicals, absolute value, percents, word problems & solids. Work through all 50 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

50	MCQ	60 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /50

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

RF

YOUR MATH COACH

Eng. Rabaa Farrag

ACT, Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

ACT SUBJECT TEST • MATH LEVEL 1 • TEST #6

Practice Questions

50 QUESTIONS

1 Rationalize the denominator and simplify $\frac{30}{\sqrt{20}}$.

(A) 3
(B) $3\sqrt{5}$
(C) $\frac{3\sqrt{2}}{2}$
(D) $3\sqrt{2}$

2 What is the solution set for $|x-3| = 2x$?

(A) $\{-3, 3\}$
(B) $\{-3\}$
(C) $\{1\}$
(D) $\{-3, 1\}$

3 What is the slope of the line whose equation is $5x - 3y = 9$?

(A) -3
(B) $-\frac{5}{3}$
(C) $\frac{5}{3}$
(D) 5

4 What is the complete factorization of $-2x^3 + 4x^2 - 8x$?

(A) $-2x(x^2 - 2x + 4)$
(B) $-4x(x^2 - x + 2)$
(C) $-2(x^2 - 2x^2 + 4x)$
(D) $-4(x^3 - x^2 + 2x)$

5 The shape of a certain city park is a regular pentagon. The length of one side is 922 ft. What is the approximate area, in square feet, of the park?

(A) 4.0 million
(B) 2.3 million
(C) 3.6 million
(D) 1.5 million

6 Of the following, 15 percent of \$8.95 is closest to:

(A) \$1.95
(B) \$1.75
(C) \$1.50
(D) \$1.35

7 Betsy raked leaves from 9:30 a.m. to 11:00 a.m. and finished $\frac{1}{4}$ of the yard. She wants to finish raking the yard by 6:30 p.m. If she plans to rake at the same rate, what is the latest time that she can start raking the leaves again?

(A) 12:30 p.m.
(B) 2:00 p.m.
(C) 1:00 p.m.
(D) 3:30 p.m.

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8 If 0.39 is about $\frac{N}{10}$, then N is closest to which of the following?

(A) 0.4
(B) 3
(C) 4
(D) 39

9 In a class, $\frac{1}{8}$ of the boys and $\frac{1}{4}$ of the girls play the piano, and 3 times as many girls as boys play the piano. If there are 100 boys and girls in the class, how many boys play the piano?

(A) 3
(B) 5
(C) 12
(D) 24

10 In the problems below, $A \div 17 = 25$ remainder 1, and $B \div 17 = 26$. What is $B - A$?

$$17\overline{)A} = 25 \text{ remainder } 1 \qquad 17\overline{)B} = 26$$

(A) 0
(B) 1
(C) 16
(D) 18

11 A store regularly sells books at 20% off the list price. At a sale, its regular prices are reduced 10 percent. The sale price is what percent of the list price?

(A) 30%
(B) 70%
(C) 72%
(D) 79%

12 A store has square rugs in three sizes. A side of the largest rug is twice as long as a side of the middle one, and a side of the middle one is twice as long as a side of the smallest one. What is the ratio of the area that the largest rug will cover to the area that the smallest one will cover?

(A) 3 to 1
(B) 4 to 1
(C) 8 to 1
(D) 16 to 1

13 With 3 weeks remaining before the recycling van comes, Al has collected 23 more cans than Bob. If Bob is to collect more cans than Al, he must average at least how many more cans per week than Al?

(A) $7\frac{2}{3}$
(B) 8
(C) 24
(D) 69

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14 If $y = 2x - 4$, then what does $y + 2$ equal?

(A) $2x - 6$
(B) $2x - 2$
(C) $4x - 6$
(D) $4x - 2$

15 The sale of a certain product increased from 25 thousand units sold in 1992 to 5 million units sold in 1996. The number of units sold in 1996 was how many times the number sold in 1992?

(A) 5
(B) 20
(C) 200
(D) 500

16 If S is the sum of two consecutive integers, which of the following is always true?

(A) S is positive
(B) S is negative
(C) S is even
(D) S is odd

17 A string is used to show where the floor of a rectangular patio is to be laid. The width of the patio is to be half the length. If 12 meters of string just fit around the edges of the floor, what is the length of the patio?

(A) 6 m
(B) 4 m
(C) 3 m
(D) 2 m

18 Each of the x members in the AZ club may bring up to 4 guests to a party. What is the maximum number of members and guests who might attend the party?

(A) $x + 4$
(B) $4x$
(C) $4x + 4$
(D) $5x$

19 Triangle A is similar to Triangle B. If Triangle B has angles measuring $30°$ and $70°$, which of the following is the measure of one of the angles of Triangle A?

(A) $20°$
(B) $40°$
(C) $60°$
(D) $80°$

20 Ann's stock was worth $28\frac{1}{4}$ a share when the stock market opened for the day and its value went down $\frac{3}{8}$ during the day. If the amounts are in dollars, how many dollars were her 40 shares of stock worth at the end of the day?

(A) $1{,}080\frac{7}{8}$
(B) $1{,}115$
(C) $1{,}085$
(D) $1{,}125$

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21 What is the slope of $3y - 6x = 10$?

(A) $\frac{1}{2}$
(B) 2
(C) 3
(D) 6

22 How many numbers between 1 and 80 (inclusive) meet both conditions below? Statement 1: The number has a factor of 4. Statement 2: The number is a multiple of 10.

(A) 2
(B) 6
(C) 4
(D) 8

23 Given that $x \geq -2$, which of the following expressions is equivalent to the product of $\sqrt{x+2}$ and $\sqrt{(x+2)^3}$?

(A) $\pm(x+2)^{\frac{3}{4}}$
(B) $x+2$
(C) $(x+2)^2$
(D) $(x+2)^4$

24 A polynomial function is defined by the equation $p(x) = 2x^2 + ax - 4x - 2a$. If $p(-3) = 0$, what is the value of a?

(A) 0
(B) 2
(C) 4
(D) 6

25 If $f(x) = 9^{2x-1}$ and $g(x) = 3^{3x+3}$, for what value of x does $f(x) - g(x) = 0$?

(A) -4
(B) $-\frac{7}{4}$
(C) $-\frac{10}{7}$
(D) 5

26 From 1970 through 1980, the population of City Q increased by 20%. From 1980 through 1990, the population increased by 30%. What was the combined percent increase for the period $1970–1990$?

(A) 25%
(B) 50%
(C) 36%
(D) 56%

27 In complex numbers, where $i^2 = -1$, what is the simplified form of the expression $\frac{(i+1)(i+1)}{(i-1)(i-1)}$?

(A) $\frac{i+1}{i-1}$
(B) $\frac{i}{2}$
(C) $\frac{2}{i}$
(D) -1

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28 The expression $x^2 + 6x + m$ can be factored as $(x + k)^2$ for some integer value of k. What is the value of m?

(A) 1
(B) 3
(C) 9
(D) 36

29 The line $y = x + 5$ undergoes a transformation such that every point (a, b) is now $(a, b + 3)$ on the resulting line. What is the equation of the line after the transformation?

(A) $y = 3x + 5$
(B) $y = 3x + 8$
(C) $y = x + 2$
(D) $y = x + 8$

30 An online candle business, open five days a week (Monday through Friday), sells fifteen candles a day its first week. If each week it sells three more candles a day, how many candles in total does that business sell in week 5?

(A) 24 candles
(B) 120 candles
(C) 27 candles
(D) 135 candles

31 How many points of intersection do the following equations have? $y = -\frac{1}{3}x + 5$ and $y = -x^2 + 8$

(A) 0
(B) 1
(C) 2
(D) 3

32 If $-3x + 7 < 16$, which of the following must be true?

(A) $x > -3$
(B) $x < -3$
(C) $x < -9$
(D) $x > -9$

33 What is the surface area of a sphere whose largest cross section has an area that measures 16π units^2?

(A) 64π units^2
(B) $\frac{256\pi}{3}$ units^2
(C) $1{,}024\pi$ units^2
(D) It is impossible to determine from the given information.

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34 Anja decides to track how long it takes her to get to school each day. After tracking her drive five times, Anja finds that it takes her an average of 27 minutes to get to school each day. The quickest Anja can get to school, if she encounters no traffic, is 23 minutes. If on the first day Anja drove to school in 31 minutes, what is the longest possible time it took Anja to drive to school during those five days?

A) 29 minutes
B) 31 minutes
C) 33 minutes
D) 35 minutes

35 What are the solutions to the equation $5|x-2|+1=16$?

A) $x=3$ only
B) $x=5$ only
C) $x=5$ and $x=3$
D) $x=5$ and $x=-1$

36 The amount of Pollutant A in a lake increases linearly over time, at a rate given by the equation $y=\frac{3}{2}x+221$, where x is time in days and y is units of Pollutant A. Pollutant B, on the other hand, is decreasing at the same rate that Pollutant A is increasing. If on Day 0 there were 108 units of Pollutant B in the lake, which equation represents the amount of Pollutant B in the lake after x days?

A) $y=\frac{2}{3}x+221$
B) $y=-\frac{2}{3}x+108$
C) $y=-\frac{3}{2}x+108$
D) $y=-\frac{3}{2}x-113$

37 Which of the following is equivalent to $2\tan^2 x\cos^2 x$?

A) $4\sin^2 x$
B) $2-\sin^2 x$
C) $2(1-\cos^2 x)$
D) $\frac{2}{\cot x\sec x}$

38 Which of the following is NOT an x-intercept of $y=x^3-4x^2-21x$?

A) $(-7,0)$
B) $(-3,0)$
C) $(0,0)$
D) $(7,0)$

39 A line is tangent to the circle $25=(x+1)^2+(y+4)^2$ at $(-4,0)$. What is the slope of that line?

A) $-\frac{4}{3}$
B) $\frac{1}{4}$
C) $\frac{3}{5}$
D) $\frac{3}{4}$

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40 What is the solution set for the equation $\left|\frac{2-x}{7}\right| = \left|\frac{x+3}{5}\right|$?

A $\left\{\frac{11}{12}\right\}$
B $\left\{-\frac{11}{12}\right\}$
C $\left\{-\frac{11}{12}, \frac{11}{6}\right\}$
D $\left\{-\frac{11}{12}, -\frac{31}{2}\right\}$

41 These tables show some ordered pairs from two linear equations. What is the x-coordinate of the solution to the system of equations?

Equation 1: x	-3	-2	0	2
y	9	7	3	-1
Equation 2: x	-2	-1	1	3
y	7	2	-8	-18

A -2
B -1
C 2
D 7

42 If $\sqrt{w}$ is greater than 8, which of the following could be the value of w?

A 9
B 16
C 49
D 81

43 Which of the following pairs of lines are parallel?

A $3y = 2x - 1$ and $y = -\frac{3}{2}x + 1$
B $y = 2x + 1$ and $y = 1 - 2x$
C $\frac{1}{2}x - y = 2$ and $2y = x - 1$
D $2y = 5x$ and $5y = 2x$

44 For all m and n, $-m(n-m) + n(m-n) =$

A $m^2 + n^2$
B $(m-n)^2$
C $(m+n)^2$
D $m^2 - n^2$

45 Two sides of a triangle have lengths 7 and 2. Which of the following could be the perimeter of this triangle?

A 9
B 11
C 14
D 16

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46 A plane intersects a hexagonal prism. The plane is perpendicular to the base of the prism. Which two-dimensional figure is the cross-section of the plane intersecting the prism?

(A) Triangle
(B) Trapezoid
(C) Hexagon
(D) Rectangle

47 A water cup in the shape of a cone has a height of 4 inches and a maximum diameter of 3 inches. What is the volume of the water in the cup, to the nearest tenth of a cubic inch, when the cup is filled to half its height?

(A) 1.2
(B) 3.5
(C) 4.7
(D) 14.1

48 A regular pyramid has a square base. The perimeter of the base is 36 inches and the height of the pyramid is 15 inches. What is the volume of the pyramid in cubic inches?

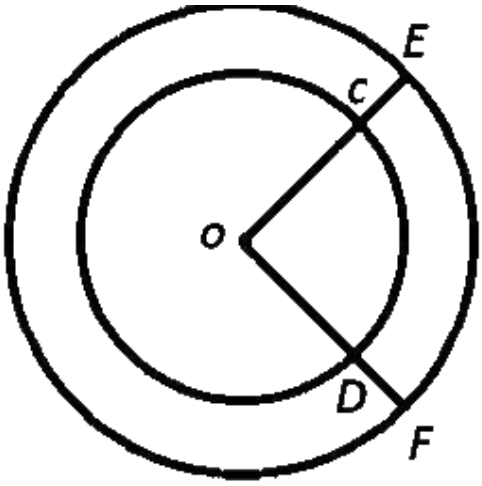
(A) 180
(B) 405
(C) 540
(D) 1215

49 A student is constructing a triangle for a geometry assignment. The teacher has specified these requirements: the perimeter of the triangle is 30 cm; the lengths of all 3 sides are whole numbers of centimeters; the shortest side of the triangle is 7 cm. How many different lengths are possible for the longest side of the triangle?

(A) 2
(B) 3
(C) 4
(D) 5

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50 In the diagram below, two concentric circles with center O, and radii $\overline{OC}$, $\overline{OD}$, $\overline{OCE}$, and $\overline{ODF}$ are drawn. If $OC = 4$ and $OE = 6$, which relationship between the length of arc EF and the length of arc CD is always true?



A The length of arc EF is 2 units longer than the length of arc CD.

B The length of arc EF is 4 units longer than the length of arc CD.

C The length of arc EF is 1.5 times the length of arc CD.

D The length of arc EF is 2.0 times the length of arc CD.

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TEST #6 • CHECK YOUR WORK

Answer Key

50 ANSWERS

Q	ANSWER	Q	ANSWER
1	B	26	D
2	C	27	D
3	C	28	C
4	A	29	D
5	D	30	D
6	D	31	C
7	B	32	A
8	C	33	A
9	B	34	D
10	C	35	D
11	C	36	C
12	D	37	C
13	B	38	A
14	B	39	D
15	C	40	D
16	D	41	A
17	B	42	D
18	D	43	C
19	D	44	D
20	B	45	D
21	B	46	D
22	C	47	A
23	C	48	B
24	D	49	B
25	D	50	C

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TEST #6 • STEP-BY-STEP

Detailed Solutions

50 SOLUTIONS

1 **Answer: B** $3\sqrt{5}$

Step 1: Simplify the root: $\sqrt{20} = 2\sqrt{5}$, so $\frac{30}{2\sqrt{5}} = \frac{15}{\sqrt{5}}$.
Step 2: Rationalize: $\frac{15}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{15\sqrt{5}}{5} = 3\sqrt{5}$.

2 **Answer: C** $\{1\}$

Step 1: An absolute value is never negative, so $2x \geq 0$, i.e. $x \geq 0$.
Step 2: Case 1: $x - 3 = 2x \Rightarrow x = -3$ — rejected (x must be ≥ 0).
Step 3: Case 2: $x - 3 = -2x \Rightarrow 3x = 3 \Rightarrow x = 1$. Check: $|1 - 3| = 2 = 2(1)$ ✓. Solution set $\{1\}$.

3 **Answer: C** $\frac{5}{3}$

Step 1: Solve for y: $-3y = -5x + 9 \Rightarrow y = \frac{5}{3}x - 3$.
Step 2: The slope is $\frac{5}{3}$.

4 **Answer: A** $-2x(x^2 - 2x + 4)$

Step 1: Each term has a factor $-2x$: $-2x^3 + 4x^2 - 8x = -2x(x^2 - 2x + 4)$.
Step 2: $x^2 - 2x + 4$ has discriminant $4 - 16 < 0$, so it cannot be factored further.

5 **Answer: D** 1.5 million

Step 1: A regular pentagon splits into 5 congruent triangles from the center; area $= \frac{5s^2}{4\tan 36^\circ}$.
Step 2: $\tan 36^\circ \approx 0.7265$, so $A \approx \frac{5(922)^2}{4(0.7265)} = \frac{4{,}250{,}420}{2.906} \approx 1{,}462{,}000$ ft^2.
Step 3: About 1.5 million square feet.

6 **Answer: D** $1.35

Step 1: $15\% = 0.15$.
Step 2: $0.15 \times 8.95 = 1.3425$.
Step 3: Closest choice: $1.35.

7 **Answer: B** 2:00 p.m.

Step 1: From 9:30 to 11:00 is 1.5 hours for $\frac{1}{4}$ of the yard.
Step 2: The remaining $\frac{3}{4}$ takes $3 \times 1.5 = 4.5$ hours.
Step 3: Latest start: 6:30 p.m. $-4{:}30 = 2{:}00$ p.m.

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8 **Answer: C** 4

Step 1: $\frac{N}{10} \approx 0.39 \Rightarrow N \approx 3.9$.
Step 2: The closest choice is 4.

9 **Answer: B** 5

Step 1: Let b = boys, so girls $= 100 - b$.
Step 2: Girl pianists $= 3\times$ boy pianists: $\frac{1}{4}(100 - b) = 3 \cdot \frac{b}{8}$.
Step 3: Multiply by 8: $200 - 2b = 3b \Rightarrow b = 40$. Boys who play: $\frac{40}{8} = 5$.

10 **Answer: C** 16

Step 1: Division check: dividend $=$ divisor $\times$ quotient $+$ remainder.
Step 2: $A = 17 \times 25 + 1 = 426$ and $B = 17 \times 26 = 442$.
Step 3: $B - A = 442 - 426 = 16$.

11 **Answer: C** 72%

Step 1: Regular price $= 80\%$ of list $= 0.80L$.
Step 2: Sale price $= 90\%$ of that $= 0.90 \times 0.80L = 0.72L$.
Step 3: The sale price is 72% of the list price.

12 **Answer: D** 16 to 1

Step 1: If the smallest side is s, the middle is $2s$ and the largest is $4s$.
Step 2: Areas: $(4s)^2 = 16s^2$ and s^2.
Step 3: Ratio $16 : 1$.

13 **Answer: B** 8

Step 1: Bob must make up 23 cans and get ahead, i.e. collect more than 23 extra cans in 3 weeks.
Step 2: Per week: more than $\frac{23}{3} = 7\frac{2}{3}$ cans.
Step 3: Cans are whole numbers, so at least 8 more cans per week.

14 **Answer: B** $2x - 2$

Step 1: Add 2 to both sides of $y = 2x - 4$.
Step 2: $y + 2 = 2x - 2$.

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15 **Answer: C** 200

Step 1: 5 million $= 5{,}000{,}000$ and 25 thousand $= 25{,}000$.
Step 2: $\frac{5{,}000{,}000}{25{,}000} = 200$ times.

16 **Answer: D** S is odd

Step 1: Two consecutive integers: n and $n+1$.
Step 2: $S = 2n+1$ — always odd. (It can be positive or negative, e.g. $-3+(-2) = -5$.)

17 **Answer: B** 4 m

Step 1: Let length $= L$, width $= \frac{L}{2}$.
Step 2: The string is the perimeter: $2\left(L + \frac{L}{2}\right) = 3L = 12$.
Step 3: $L = 4$ m.

18 **Answer: D** $5x$

Step 1: Each member can bring up to 4 guests, so each member accounts for up to $1+4=5$ people.
Step 2: With x members: at most $5x$ people.

19 **Answer: D** $80°$

Step 1: Similar triangles have the same three angles.
Step 2: Triangle B's third angle: $180° - 30° - 70° = 80°$.
Step 3: So Triangle A has angles $30°$, $70°$, $80°$ — $80°$ is the only listed choice.

20 **Answer: B** 1,115

Step 1: End-of-day price: $28\frac{1}{4} - \frac{3}{8} = 28.25 - 0.375 = 27.875$.
Step 2: 40 shares: $40 \times 27.875 = 1115$ dollars.

21 **Answer: B** 2

Step 1: Solve for y: $3y = 6x + 10 \Rightarrow y = 2x + \frac{10}{3}$.
Step 2: The slope is 2.

22 **Answer: C** 4

Step 1: "Has a factor of 4" and "multiple of 10" means a multiple of $\text{lcm}(4, 10) = 20$.
Step 2: Multiples of 20 from 1 to 80: $20, 40, 60, 80$.
Step 3: 4 numbers.

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23 **Answer: C** $(x+2)^2$

Step 1: Write as powers: $\sqrt{x+2} = (x+2)^{\frac{1}{2}}$ and $\sqrt{(x+2)^3} = (x+2)^{\frac{3}{2}}$.
Step 2: Add exponents: $(x+2)^{\frac{1}{2}+\frac{3}{2}} = (x+2)^2$.

24 **Answer: D** 6

Step 1: Substitute $x = -3$: $2(9) + a(-3) - 4(-3) - 2a = 0$.
Step 2: $18 - 3a + 12 - 2a = 30 - 5a = 0$.
Step 3: $a = 6$.

25 **Answer: D** 5

Step 1: Write both with base 3: $9^{2x-1} = (3^2)^{2x-1} = 3^{4x-2}$.
Step 2: $f(x) = g(x) \Rightarrow 3^{4x-2} = 3^{3x+3} \Rightarrow 4x - 2 = 3x + 3$.
Step 3: $x = 5$.

26 **Answer: D** 56%

Step 1: Percent increases multiply: growth factors 1.20 and 1.30.
Step 2: $1.20 \times 1.30 = 1.56$.
Step 3: Combined increase: 56%.

27 **Answer: D** -1

Step 1: $(i+1)^2 = i^2 + 2i + 1 = -1 + 2i + 1 = 2i$.
Step 2: $(i-1)^2 = i^2 - 2i + 1 = -2i$.
Step 3: $\frac{2i}{-2i} = -1$.

28 **Answer: C** 9

Step 1: $(x+k)^2 = x^2 + 2kx + k^2$.
Step 2: Match: $2k = 6 \Rightarrow k = 3$.
Step 3: $m = k^2 = 9$.

29 **Answer: D** $y = x + 8$

Step 1: $(a, b) \to (a, b+3)$ moves every point up 3 units.
Step 2: $y = x + 5$ shifted up 3: $y = x + 8$.

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30 **Answer: D** 135 candles

Step 1: Candles per day: week 1: 15; each week adds 3, so week 5: $15 + 4 \times 3 = 27$.
Step 2: Five days: $27 \times 5 = 135$ candles.

31 **Answer: C** 2

Step 1: Set equal: $-x^2 + 8 = -\frac{1}{3}x + 5$.
Step 2: $x^2 - \frac{1}{3}x - 3 = 0$; discriminant $= \left(\frac{1}{3}\right)^2 + 12 > 0$.
Step 3: Two real solutions, so 2 intersection points.

32 **Answer: A** $x > -3$

Step 1: Subtract 7: $-3x < 9$.
Step 2: Divide by -3 and flip the sign: $x > -3$.

33 **Answer: A** 64π units^2

Step 1: The largest cross section of a sphere is a great circle: $\pi r^2 = 16\pi \Rightarrow r = 4$.
Step 2: Surface area $= 4\pi r^2 = 4\pi(16) = 64\pi$ units^2.

34 **Answer: D** 35 minutes

Step 1: Average 27 over 5 days: total $= 135$ minutes.
Step 2: Day 1 was 31, leaving 104 for the other four days.
Step 3: To make one day as long as possible, make the other three as short as possible (23 each): $104 - 69 = 35$ minutes.

35 **Answer: D** $x = 5$ and $x = -1$

Step 1: Isolate: $5|x - 2| = 15 \Rightarrow |x - 2| = 3$.
Step 2: $x - 2 = 3 \Rightarrow x = 5$; $x - 2 = -3 \Rightarrow x = -1$.

36 **Answer: C** $y = -\frac{3}{2}x + 108$

Step 1: Pollutant A increases at $\frac{3}{2}$ units per day, so B decreases at $\frac{3}{2}$ per day: slope $-\frac{3}{2}$.
Step 2: Starting amount on day 0: 108.
Step 3: $y = -\frac{3}{2}x + 108$.

37 **Answer: C** $2(1 - \cos^2 x)$

Step 1: $\tan^2 x = \frac{\sin^2 x}{\cos^2 x}$, so $2\tan^2 x \cos^2 x = 2\sin^2 x$.
Step 2: Using $\sin^2 x = 1 - \cos^2 x$: $2(1 - \cos^2 x)$.

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38 **Answer: A** $(-7, 0)$

Step 1: Factor: $x^3 - 4x^2 - 21x = x(x^2 - 4x - 21) = x(x - 7)(x + 3)$.
Step 2: x-intercepts: $(0, 0)$, $(7, 0)$, $(-3, 0)$.
Step 3: $(-7, 0)$ is not an x-intercept.

39 **Answer: D** $\frac{3}{4}$

Step 1: The circle's center is $(-1, -4)$.
Step 2: Slope of the radius to $(-4, 0)$: $\frac{0-(-4)}{-4-(-1)} = \frac{4}{-3} = -\frac{4}{3}$.
Step 3: A tangent is perpendicular to the radius: slope $= \frac{3}{4}$.

40 **Answer: D** $\left\{-\frac{11}{12}, -\frac{31}{2}\right\}$

Step 1: $\left|\frac{2-x}{7}\right| = \left|\frac{x+3}{5}\right|$ means $\frac{2-x}{7} = \pm\frac{x+3}{5}$.
Step 2: Plus: $5(2 - x) = 7(x + 3) \Rightarrow 10 - 5x = 7x + 21 \Rightarrow x = -\frac{11}{12}$.
Step 3: Minus: $5(2 - x) = -7(x + 3) \Rightarrow 10 - 5x = -7x - 21 \Rightarrow x = -\frac{31}{2}$.

41 **Answer: A** -2

Step 1: Equation 1: slope $= \frac{7-9}{-2+3} = -2$, through $(0, 3)$: $y = -2x + 3$.
Step 2: Equation 2: slope $= \frac{2-7}{-1+2} = -5$, through $(-2, 7)$: $y = -5x - 3$.
Step 3: $-2x + 3 = -5x - 3 \Rightarrow x = -2$ (both tables contain $(-2, 7)$).

42 **Answer: D** 81

Step 1: Square both sides: $\sqrt{w} > 8 \Rightarrow w > 64$.
Step 2: Only 81 is greater than 64.

43 **Answer: C** $\frac{1}{2}x - y = 2$ and $2y = x - 1$

Step 1: Parallel lines have equal slopes (and different intercepts). Put each in $y = mx + b$ form.
Step 2: C: $\frac{1}{2}x - y = 2 \Rightarrow y = \frac{1}{2}x - 2$, and $2y = x - 1 \Rightarrow y = \frac{1}{2}x - \frac{1}{2}$.
Step 3: Both slopes are $\frac{1}{2}$, so the lines are parallel.

44 **Answer: D** $m^2 - n^2$

Step 1: Distribute: $-mn + m^2 + mn - n^2$.
Step 2: $-mn$ and $+mn$ cancel: $m^2 - n^2$.

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45 **Answer: D** 16

Step 1: Triangle inequality: third side c satisfies $7-2<c<7+2$, i.e. $5<c<9$.
Step 2: Perimeter $=9+c$ is between 14 and 18 (not including them).
Step 3: Only 16 fits.

46 **Answer: D** Rectangle

Step 1: A hexagonal prism has rectangular side faces standing on the hexagonal base.
Step 2: A plane perpendicular to the base cuts these faces in vertical segments of the prism's height, with horizontal edges on the top and bottom faces.
Step 3: The cross section is a rectangle.

47 **Answer: A** 1.2

Step 1: At half the height $(2$ in), similar triangles give half the radius: $\frac{1.5}{2}=0.75$ in.
Step 2: Water volume $=\frac{1}{3}\pi r^2 h=\frac{1}{3}\pi(0.75)^2(2)$.
Step 3: $\approx 1.18\approx 1.2$ in^3.

48 **Answer: B** 405

Step 1: Square base with perimeter 36: side $=9$, base area $=81$.
Step 2: $V=\frac{1}{3}Bh=\frac{1}{3}(81)(15)=405$ in^3.

49 **Answer: B** 3

Step 1: Let the other sides be $a\le b$: $7+a+b=30\Rightarrow a+b=23$, with $a\ge 7$.
Step 2: Triangle inequality: $b<7+a=7+(23-b)\Rightarrow 2b<30\Rightarrow b<15$. Also $b\ge a\Rightarrow b\ge 11.5$.
Step 3: b can be 12, 13 or 14: 3 possible lengths.

50 **Answer: C** The length of arc EF is 1.5 times the length of arc CD.

Step 1: Both arcs are cut off by the same central angle, so each arc is the same fraction of its circle.
Step 2: Arc length is proportional to the radius: $\frac{EF}{CD}=\frac{OE}{OC}=\frac{6}{4}=1.5$.
Step 3: Arc EF is 1.5 times arc CD.

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